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Retrospective Evaluation of Venous Thromboembolism Prophylaxis in Elderly, High-Risk Trauma Patients

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ABSTRACT

Background: Venous thromboembolism (VTE) risk increases with age. Scarce data exist for patients age ≥65 y. This study evaluated VTE incidence in elderly, high-risk trauma patients receiving unfractionated heparin (UFH) or enoxaparin chemoprophylaxis.

Materials and methods: This retrospective, single-center, cohort study included trauma patients age ≥ 65 y with risk assessment profile (RAP) ≥ 5 who received UFH or enoxaparin chemoprophylaxis. The primary outcome was VTE incidence requiring therapeutic anticoagulation. An age-modified RAP (RAP-AM) was calculated as RAP without age distribution points. Logistic regression analyses were performed to identify independent predictors for VTE development and chemoprophylactic agent selection. Bleeding incidence compared packed red blood cells utilized.

Results: A total of 1090 patients were included (UFH, $n = 655$; enoxaparin, $n = 435$). VTE occurred in 39 (3.6%) patients with no difference between groups in proximal deep vein thrombosis (2.1% *versus* 3.0%, $P = 0.52$) or pulmonary embolism (1.2% *versus* 1.4%, $P = 0.96$). Weight ≥125 kg (OR 4.12, 95% CI 1.06-16.11) and RAP-AM ≥ 5 (OR 6.52, 95% CI 2.65-16.03) were independently associated with VTE development. Increasing age (OR 1.04, 95% CI 1.03-1.06), initiation ≤ 24 h (OR 2.17, 95% CI 1.66-2.84) and creatinine clearance ≤ 30 mL/min (OR 1.61, 95% CI 1.17-2.21) were independent predictors of receiving UFH whereas increasing ISS (OR 0.97, 95% CI 0.95-0.99) was associated with receiving enoxaparin.

Conclusions: VTE incidence may be similar for high-risk, elderly trauma patients receiving UFH and enoxaparin chemoprophylaxis. Further research is necessary to determine noninferiority of UFH to enoxaparin in this patient population.

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Introduction

Venous thromboembolism (VTE) is associated with high morbidity, mortality, and immense economic burden.[1] Trauma patients have an increased VTE risk with a reported incidence as high as 57% without chemoprophylaxis.[2-7] Chemoprophylaxis selections consist of low-dose unfractionated heparin (UFH), low-molecular-weight heparin (LMWH), or mechanical prophylaxis.[8,9] Chemoprophylaxis studies and the Eastern Association for the Surgery of Trauma guidelines for prevention of VTE suggest moderate- to high-risk patients (e.g., spinal cord injury, pelvic fractures, Injury Severity Score [ISS] $\geq$ 9) should receive LMWH for VTE prevention.[5,10-16]

Increasing age has been reported as an independent risk factor for VTE, but data are limited in patients with advanced age.[4,16-18] Although patients age 65 y and older account for nearly 20% of the trauma population, elderly patients are largely underrepresented in VTE chemoprophylaxis studies wherein the average age of included patients is roughly 44 y.[4,5,16,17,19] Recent data from the National Trauma Database demonstrated increased VTE risk up to age 65 y, but age $\geq$ 65 y was negatively associated with VTE, suggesting that advanced age may have a diminishing relationship with VTE risk.[20] Moreover, reduced drug elimination in elderly patients may cause LMWH accumulation resulting in bleeding, potentially limiting the safety of these agents in patients with advanced age.[21-23] Thus, when weighing the risk *versus* benefit for the ideal chemoprophylaxis agent, elderly patients may represent a clinical conundrum to safely and effectively prevent VTE given the paucity of data in this growing age-based subgroup of trauma patients.[3,4,6,20,24-26]

The purpose of the study was to evaluate the effectiveness of UFH *versus* enoxaparin for VTE prevention in elderly, high-risk trauma patients. We hypothesized that VTE incidence would be higher for patients receiving UFH *versus* enoxaparin.

Materials and methods

Study design and patient cohorts

This retrospective, single-center, cohort study included trauma patients admitted to the University of Cincinnati Medical Center (UCMC), an urban American College of Surgeons–verified level 1 trauma center, between January 1, 2013, and December 31, 2017. The study was approved and conducted in accordance with the University of Cincinnati Investigational Review Board. Patients age 65 y and older with a Greenfield Risk Assessment Profile (RAP)[17] $\geq$ 5 were identified from the local trauma registry and classified into two groups based on the initial chemoprophylactic agent administered: UFH or enoxaparin. Patients were excluded if VTE was diagnosed before chemoprophylaxis initiation, therapeutic anticoagulation was recorded as the first administered agent, or if they were actively incarcerated.

Chemoprophylaxis agent selection was guided by the VTE prevention protocol at UCMC, modified from the original protocol.[24] In brief, the protocol stated that high-risk patients (i.e., RAP $\geq$ 5) were to receive enoxaparin 30 mg subcutaneously every 12 h; however, patient-specific alterations could be made at the discretion of the attending trauma physician. Patients receiving enoxaparin had serum anti-factor Xa concentrations (anti-Xa) drawn 30 min before the fourth dose. If anti-Xa was <0.1 IU/mL, the enoxaparin dose was adjusted to 40 mg every 12 h. Obese high-risk patients, defined as absolute body weight $\geq$125 kg or body mass index $\geq$40, were started on enoxaparin 40 mg every 12 h with adjustment to 50 mg every 12 h if anti-Xa < 0.1 IU/mL. Anti-Xa concentrations were not obtained in patients receiving UFH. Bilateral lower-extremity compression devices were applied to all patients on admission unless contraindicated. High-risk patients had surveillance duplex ultrasound of the bilateral lower extremities per standard of care performed on post-injury day 3 and then every 7 d during admission thereafter if admitted between Winter 2013 and Fall 2016. The screening duplex protocol was modified Fall 2016 to be performed on patients with RAP $\geq$8 on admission, hospital day 7, then weekly thereafter while inpatient. The trauma attending could elect not to perform routine VTE screening. VTE was clinically diagnosed by signs and symptoms with confirmation via duplex ultrasound or computed tomography pulmonary angiography, as appropriate.

Data collection and outcomes

Demographic data including age, sex, weight, body mass index, injury mechanism (e.g., blunt or penetrating), ISS, RAP, packed red blood cell (PRBC) transfusion requirements, intensive care unit and hospital length of stay, and admission serum creatinine were obtained through the UCMC trauma registry and commercial electronic medical record. The electronic medical record programming empirically rounded serum creatinine values < 1.0 mg/dL to 1.0 mg/dL in patients aged $\geq$65 y for estimation of creatinine clearance using the Cockcroft-Gault formula, which were used for this study.[25] Age-modified RAP (RAP-AM) was calculated by removing age points (40-59 y, 2 points; 60-74 y, 3 points; $\geq$ 75 y, 4 points) to isolate the impact of underlying conditions, iatrogenic factors, and injury-related factors on VTE risk.

The primary outcome was incidence of in-hospital VTE. VTE was defined as proximal deep vein thrombosis (DVT), pulmonary embolism (PE), or other clinically relevant VTE resulting in therapeutic treatment per medical team decision. All VTE events were identified through an internal trauma registry and confirmed by manual electronic medical record review. Secondary outcomes included identifying the association of RAP, RAP-AM, and ISS with VTE occurrence, VTE risk factors, predictors for prescribing UFH and enoxaparin, and bleeding incidence. Bleeding was defined as the administration of two or more units of PRBC within a 24-h period. Bleeding events were compared between groups at two time points: (1) the first 24 h of admission or (2) after the first 24 h of admission until discharge.

Statistical analysis

A convenience sample over a 5-y period was used for sample size determination to promote broader inclusion of the proposed subset of advanced age, high-risk trauma patients managed during a consistent period. As such, all patients identified from the trauma registry admitted within the 5-y period were assessed for inclusion.

Patient demographics and outcome information were compared between groups. Chi-squared or Fisher exact test was used for categorical data and Student's *t*-test or Wilcoxon rank-sum was used for continuous data, as appropriate. Statistical significance was set at a *P*-value < 0.05. Receiver operating characteristic (ROC) curves were constructed to assess RAP, RAP-AM, and ISS ability to predict VTE. ROC curve inflection points were identified by finding sensitivity of 90% then adjusting sensitivity to maximize corresponding specificity.

Independent risk factors for VTE were assessed by multivariate logistic regression (MVLR). A second MVLR analysis was performed to identify clinical and demographic factors associated with receiving UFH or enoxaparin across the cohort. Variables with *P*-value < 0.2 on univariate analysis between respective cohorts were included in the MVLR. Unfractionated heparin use was assessed in the VTE risk model, and RAP-AM breakpoint identified via ROC curve was evaluated for inclusion, as described. ISS $\geq$9, weight $\geq$125 kg, age $\geq$70 y, age $\geq$75 y, age $\geq$80 y, and age $\geq$85 y were each evaluated *post hoc* in the VTE risk model. ISS $\geq$ 9 was assessed instead of the identified breakpoint on ROC as prior literature used this threshold to define high-risk.[5] Creatinine clearance $\leq$30 mL/min and RAP break point identified via ROC curve were evaluated for inclusion in the model evaluating UFH use. Goodness of fit was described using the Hosmer–Lemeshow statistic. The best-fit model for each MVLR analysis was presented. Statistical analyses were performed using SigmaPlot version 13.0 (Systat Software, San Jose, CA).

Results

A total of 1237 patients admitted to the trauma service with an age $\geq$65 y and RAP $\geq$5 were screened (Fig. 1). One thousand and ninety (UFH, $n = 655$; enoxaparin, $n = 435$) were included. Baseline demographics are shown in Table 1. The overall median age was 76 (IQR 70-83) y, median RAP 8 (6-10) and median ISS 14 (9-17). Patients receiving UFH were older (77 *versus* 74 y, $P < 0.05$), had lower actual body weight (74.8 *versus* 78 kg, $P < 0.05$), and lower creatinine clearance (43.6 *versus* 50.4 mL/min, $P < 0.05$) compared with the enoxaparin group. In addition, ISS (12 *versus* 14, $P < 0.05$), RAP (8 *versus* 9, $P < 0.05$), and RAP-AM (4 *versus* 5, $P < 0.05$) were lower in the UFH group. Median hospital (5 *versus* 6 d, $P < 0.05$) and intensive care unit lengths of stay (3 *versus* 3 d, $P < 0.05$) were shorter in the UFH group, but in-hospital mortality was similar. Time to administration of chemoprophylaxis (19.4 *versus* 31.9 h, $P < 0.05$) was shorter in the UFH group.

Venous thromboembolism occurred in 39 (3.6%) patients overall, with 21 (3.2%) in UFH group and 18 (4.1%) in enoxaparin group ($P = 0.519$; Table 2). There were no differences in rates of individual DVT types or PE between groups; one patient in each group had both a PE and DVT. Evaluation of age classifications revealed no differences in VTE between groups (Fig. 2).

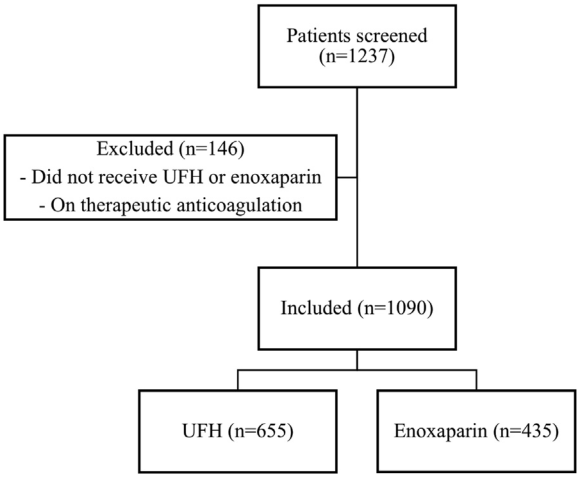

Fig. 1 – Patients selection criteria. Obtained from the trauma registry.

Venous thromboembolism prediction using ROC curve analysis demonstrated areas under the curve for ISS, RAP, and RAP-AM as 0.71, 0.77, and 0.78, respectively (Fig. 3). The optimal ROC curve inflection points were ISS $\geq$13 (90% sensitivity, 46% specificity), RAP $\geq$ 8 (86% sensitivity, 52% specificity), and RAP-AM $\geq$ 5 (85% sensitivity, 60% specificity). An ISS of $\geq$9 had a sensitivity of 94% and specificity of 21%.

Factors associated with VTE development on univariate analysis included age $\geq$85 y, female sex, weight $\geq$125 kg, ISS $\geq$9, RAP-AM $\geq$ 5, and time to chemoprophylaxis initiation. On MVLR, only weight $\geq$125 kg (OR 4.12, 95% CI 1.06-16.11; $P = 0.042$) and RAP-AM $\geq$ 5 (OR 6.52, 95% CI 2.65-16.03; $P < 0.001$) were identified as independent predictors for VTE (Fig. 4). Age, ISS, RAP $\geq$8, creatinine clearance $\leq$30 mL/min, time to chemoprophylaxis initiation, and blunt trauma were associated with receiving UFH on univariate analysis. Age (OR 1.04, 95% CI 1.03-1.06; $P < 0.001$), creatinine clearance $\leq$ 30 mL/min (OR 1.61, 95% CI 1.17-2.21; $P = 0.004$), chemoprophylaxis initiation within 24 h (OR 2.17; 95% CI 1.66-2.84; $P < 0.001$), and ISS (OR 0.97; 95% CI 0.95-0.99; $P = 0.002$) were identified on MVLR to be independently associated with receiving UFH (Fig. 5).

Bleeding incidence was similar between groups. Within the first 24 h of admission, 25 (3.5%) UFH and 20 (4.6%) enoxaparin patients required two or more units of PRBC ($P = 0.63$). Bleeding after the initial 24 h through discharge was observed in 33 (5.0%) UFH and 35 (8.0%) enoxaparin patients ($P = 0.06$).

Discussion

This retrospective cohort study of elderly, high-risk trauma patients found no statistically significant difference in VTE, DVT, PE, or bleeding outcomes between UFH and enoxaparin chemoprophylaxis. Unfractionated heparin was not associated with VTE incidence on regression analysis controlling for various VTE risk factors. Notably, patients with higher ISS and RAP were more likely to receive enoxaparin, likely influenced

Table 1 – Baseline characteristics.

Characteristic	UFH ($n = 655$)	Enoxaparin ($n = 435$)
Age, y*	77 (71-84)	74 (69-81)
Female, n (%)	324 (50)	199 (46)
Weight (kg)*	74.8 (62-89.8) ($n = 649$)	78 (66-90.9) ($n = 432$)
BMI, kg/m^2	26.1 (22.7-30.6) ($n = 645$)	26.6 (23.7-30.6) ($n = 428$)
Serum creatinine, mg/dL*,†	1.0 (1.04-1.37) ($n = 618$)	1.0 (1.0-1.20) ($n = 416$)
Creatinine clearance, mL/min*	43.6 (33.2-56.1) ($n = 646$)	50.4 (41.0-64.7) ($n = 431$)
Trauma type, blunt, n (%)	634 (96.8)	428 (98.4)
ISS*	12 (9-17)	14 (10-19)
RAP*	8 (6-10)	9 (7-11)
RAP-AM*	4 (2-6)	5 (3-7)
ICU LOS*	3 (0-5)	3 (1-6)
Hospital LOS*	5 (3-9)	6 (4-10)
Initiation time, h*	19.4 (8.16-37.4)	31.9 (23.0-47.5)
Initiation within 24 h, n (%)*	376 (57.4)	159 (36.6)
Initiation within 48 h, n (%)*	567 (86.6)	329 (75.6)
Initiation within 72 h, n (%)*	629 (96)	396 (91)
In-hospital mortality, n (%)	35 (5.3)	25 (5.7)

Data presented as median (interquartile range), unless otherwise stated.
BMI = body mass index; ICU = intensive care unit; LOS = length of stay; RAP = Risk Assessment Profile; RAP-AM = Risk Assessment Profile-Age Modified; EMR = electronic medical record.
*P-value <0.05.
†Values <1 mg/dL rounded to 1 mg/dL. With serum clearance <1 and age ≥ 65, the EMR rounds to 1 mg/dL for Cockroft-Gault calculations. Values were rounded to simulate the creatinine clearance providers would view.

by the local protocol that recommended enoxaparin for the highest-risk patients. To our knowledge, this is the first study to evaluate VTE chemoprophylaxis agent selection and VTE incidence in elderly, high-risk trauma patients. Our results support the use of either UFH or enoxaparin chemoprophylaxis in this population; however, the findings may not be applicable to the highest-risk patients as this subgroup primarily received enoxaparin.

Multiple studies describe enoxaparin superiority or UFH noninferiority for chemoprophylaxis in high-risk trauma patients. These findings may be explained by medication dose, administration frequency, young patient populations, and study design heterogeneity.[4,16,27,28] Our study population had a median age of 77 y with a range of 65-102 y, a subpopulation of trauma patients poorly represented in published trials. As evidenced by 53.8% of our high-risk, elderly patients receiving UFH as the first chemoprophylactic agent administered, concerns that these patients may have a different risk:benefit relationship relative to their younger counterparts may influence clinical practice decisions. On MVLR, the odds of

Table 2 – Venous thromboembolism incidence.

Outcome	UFH ($n = 655$)	Enoxaparin ($n = 435$)
VTE	21 (3.2)	18 (4.1)
DVT	14 (2.1)	13 (3.0)
PE	8 (1.2)	6 (1.4)
DVT and PE	1 (0.15)	1 (0.23)

Data presented as n (%).
No observations presented were statistically significant.
VTE = venous thromboembolism; DVT = deep vein thrombosis; PE = pulmonary embolism.

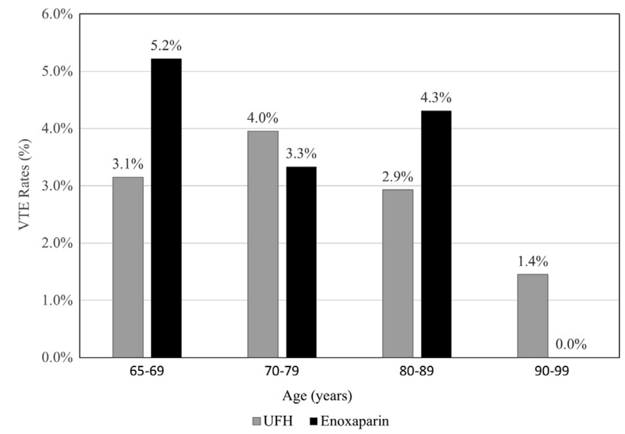

Fig. 2 – Venous thromboembolism rates between age groups. No statistically significant differences exist between enoxaparin and unfractionated heparin for any age group.

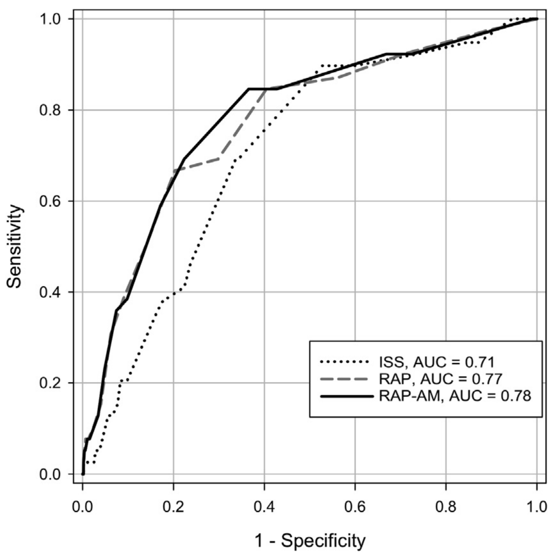

Fig. 3 – Receiver operating characteristic curve for venous thromboembolism incidence. AUC = area under the curve.

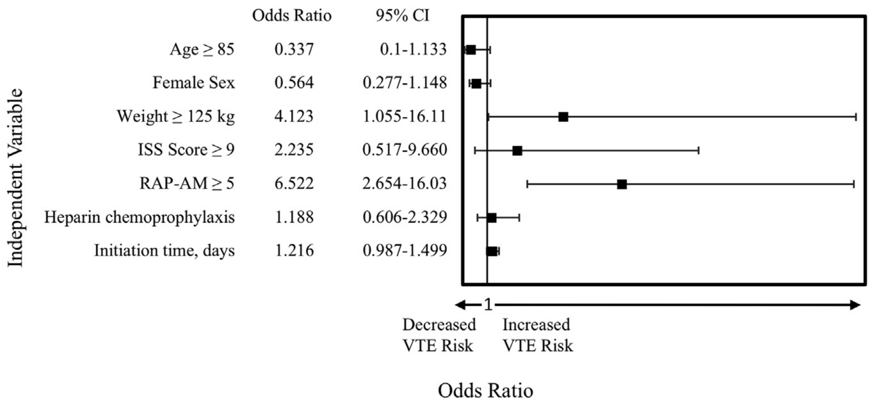

Fig. 4 – Predictors of venous thromboembolism risk. Hosmer–Lemeshow Statistic: P = 0.749. VTE = venous thromboembolism; ISS = Injury Severity Score; RAP-AM = Risk Assessment Profile-Age Modified; CI = confidence interval.

receiving UFH increased 4% with each year increase in age. This may have been further driven by decreased renal function, which was also positively associated with receiving UFH. With no difference in VTE between chemoprophylactic cohorts, nonspecifically extrapolating current literature to high-risk elderly patients may fail to consider other important factors that uniquely balance the risk:benefit relationship in this population such as age-independent VTE risk (e.g., ISS) and age-related pharmacokinetic alterations (e.g., inconsistent subcutaneous absorption; decreased total body water; and reduced renal blood flow).[29-31]

One approach to help guide optimal chemoprophylaxis in elderly, high-risk patients is appropriate VTE risk assessment. Kim *et al* reported a 3.2% VTE rate in all elderly trauma patients.[32] Nastasi *et al* found a negative association between patients age ≥65 y and VTE diagnosis rates.[20] In comparison, a general adult patient population found VTE rates of 8.2% and 5.1% for UFH and enoxaparin, respectively.[27] Our study found lower VTE rates in the high-risk elderly patient population, similar to Kim *et al*; this suggests while risk increases with age in adult trauma patients, it possibly plateaus or even decreases in patients age 65 y or older.[20,32] Although RAP includes age to assess risk of VTE, the average age of the validation cohort was 40 y, thus its utility in identifying VTE risk in elderly patients has not previously been described.[28] Removing age as a risk factor in the RAP-AM assessment allows isolated assessment of underlying condition, iatrogenic factors, and injury-related factors. Consequently, RAP-AM had the highest area under the curve for VTE prediction compared with RAP and ISS. In addition, an RAP-AM of ≥5 had a 7-times higher odds of VTE in contrast to an ISS ≥9, which was not associated with VTE. Further research should be performed to clarify the best VTE risk calculator for elderly trauma patients.

Studies show less than a 2% risk of major bleeding with prophylactic anticoagulation after trauma, and many show no statistically significant differences between UFH and enoxaparin bleeding rates.[5,27,33-35] One study reported elderly trauma patients had a 4.1% likelihood of transfusion within 24 h of injury.[36] This study found similar rates of administration of blood within the first 24 h. While a higher bleeding incidence in the enoxaparin group (5% *versus* 8%, $P = 0.06$; 37.5% relative risk reduction and 3% absolute risk reduction toward UFH) was found after 24 h of injury, dosing and serum factor anti-Xa results were not collected which would provide greater description and interpretation of this finding. Further studies should elucidate bleeding risk when juxtaposed to VTE reduction in elderly patients.

Extrapolation of the present study results is limited by the single-center, retrospective design, and use of a convenience sample. Patients were classified into groups based on first

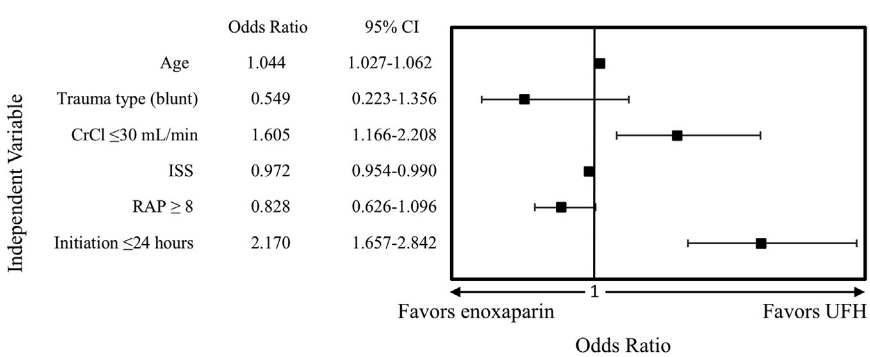

Fig. 5 – Predictors of chemoprophylaxis agent selection. Hosmer–Lemeshow Statistic: P = 0.736. CrCl = creatinine clearance; ISS = Injury Severity Score; RAP = Risk Assessment Profile; CI = confidence interval.

chemoprophylactic agent administered, leading to risk for both selection bias and misclassification for medication used throughout the admission. Confounding factors may be present given institutional protocol prescribing bias toward enoxaparin in the highest-risk patients. Further data on chemoprophylaxis dosing, missed or held doses, anti-Xa concentrations, VTE detection due to symptoms *versus* routine duplex or transition to therapeutic anticoagulation were not collected. Providers could individualize chemoprophylaxis management and suspend route duplex screening if clinically indicated; however, protocol deviation reasons were not collected. UCMC duplex compliance based on Martin *et al* research was 81% in a 49 y old median patient population.[37] No screening of therapeutic anticoagulation before admission was performed, leading to an unknown population influenced by home blood-thinning agents. The safety outcome of bleeding only assessed number of packed red blood cells to identify patients bleeding, which does not account for clinical relevance. Despite these limitations, this is the only study to our knowledge that investigated the use of UFH *versus* enoxaparin chemoprophylaxis in an elderly, high-risk trauma population.

Conclusions

This study suggests no difference in VTE incidence between high-risk elderly trauma patients receiving UFH or enoxaparin chemoprophylaxis. Future research is necessary to determine noninferiority of UFH to enoxaparin. RAP-AM may better identify VTE risk within elderly trauma patients; however, validation should be performed in all elderly patients.

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Authors' contributions: Erica N. Krantz, PharmD, conducted the literature search, study design, data collection, analysis, interpretation, and writing. Carolyn D. Philpott, PharmD, BCCCP, and Molly E. Droege, PharmD, BCPS, assisted with the literature search, study design, data analysis, interpretation, and critical review of the manuscript. Christopher A. Droege, PharmD, BCCCP, FCCM, FASHP assisted with study design, data interpretation, and critical review of the manuscript. Neil E. Ernst, PharmD, and Paige M. Garber, PharmD, BCCCP, assisted with study design and manuscript review. Betty J. Tsuei, MD, FACS, FCCM, and Michael D. Goodman, MD, FACS, assisted with data interpretation and manuscript. Eric W. Mueller, PharmD, FCCM, FCCP, assisted with statistical analysis and manuscript review.

Disclosure

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